

Translated

From: e-maxx.ru

Linear Congruence Equation

This equation is of the form:

$$a \cdot x \equiv b \pmod{n},$$

where a , b and n are given integers and x is an unknown integer.

It is required to find the value x from the interval $[0, n-1]$ (clearly, on the entire number line there can be infinitely many solutions that will differ from each other in $n \cdot k$, where k is any integer). If the solution is not unique, then we will consider how to get all the solutions.

Solution by finding the inverse element

Let us first consider a simpler case where a and n are **coprime** ($\gcd(a, n) = 1$). Then one can find the **inverse** of a , and multiplying both sides of the equation with the inverse, and we can get a **unique** solution.

$$x \equiv b \cdot a^{-1} \pmod{n}$$

Now consider the case where a and n are **not coprime** ($\gcd(a, n) \neq 1$). Then the solution will not always exist (for example $2 \cdot x \equiv 1 \pmod{4}$ has no solution).

Let $g = \gcd(a, n)$, i.e. the **greatest common divisor** of a and n (which in this case is greater than one).

Then, if b is not divisible by g , there is no solution. In fact, for any x the left side of the equation $a \cdot x \pmod{n}$, is always divisible by g , while the right-hand side is not divisible by it, hence it follows that there are no solutions.

If g divides b , then by dividing both sides of the equation by g (i.e. dividing a , b and n by g), we receive a new equation:

$$a' \cdot x \equiv b' \pmod{n'}$$

in which a' and n' are already relatively prime, and we have already learned how to handle such an equation. We get x' as solution for x .